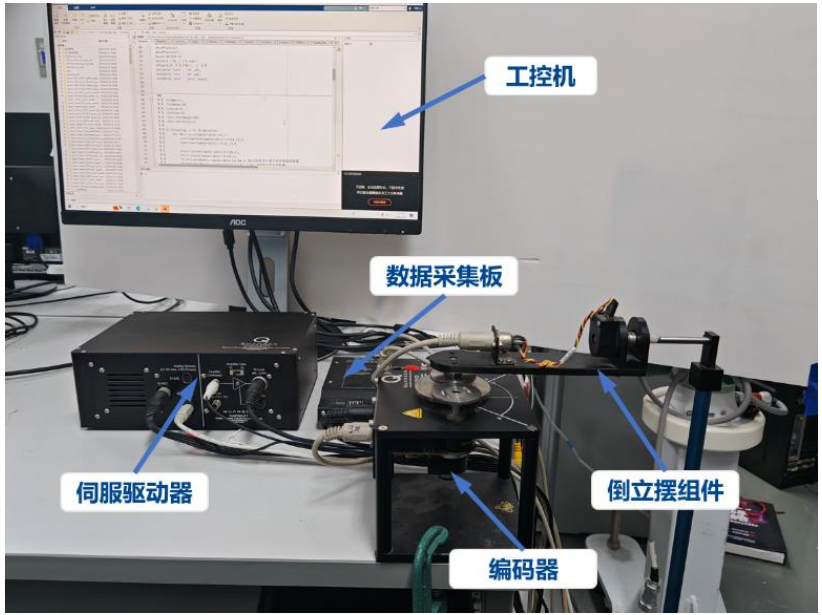
机电系统建模、辨识与控制
Modeling, Identification and Control of Mechatronic Systems

MIC_L5 System Identification—— Case Study

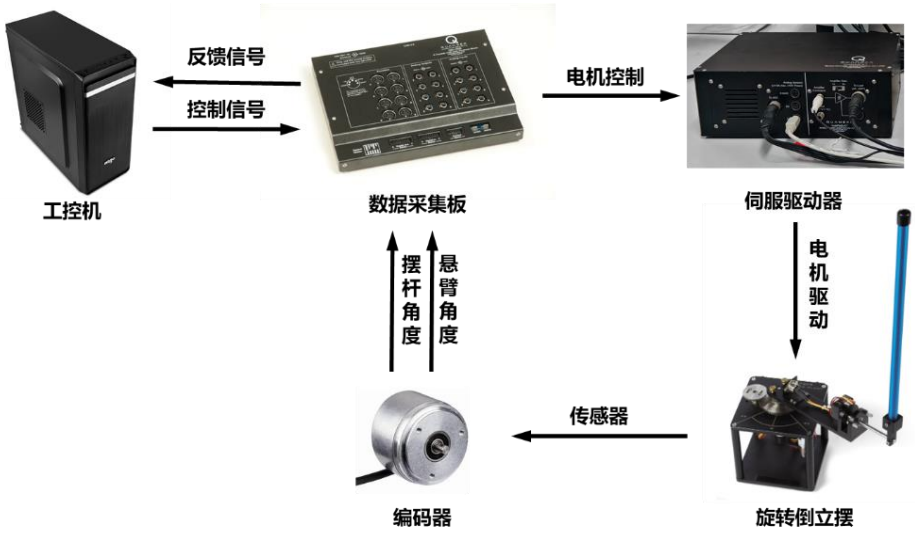
朱笑丛（Xiaocong Zhu） 副教授

Test Rig of Inverted Pendulum System

- **Test rig**



Test rig of rotary inverted pendulum system

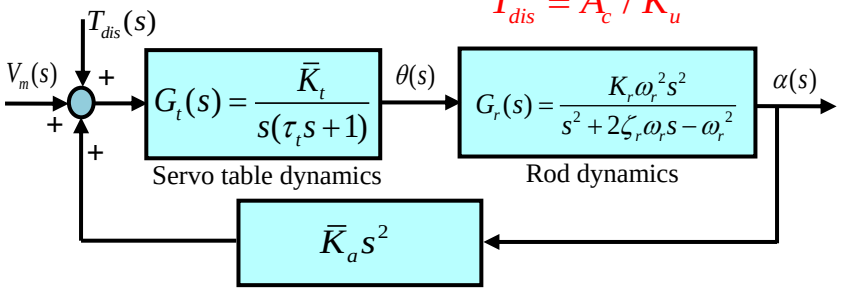


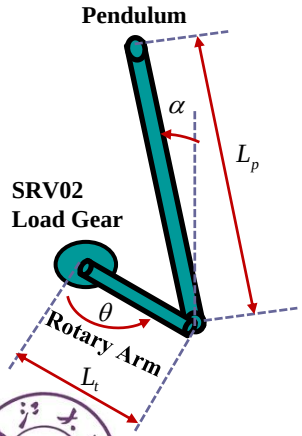
Schematics of rotary inverted pendulum system

Model of Rotary Inverted Pendulum

- **Rod upward**

$T_{dis} = A_c / K_u$





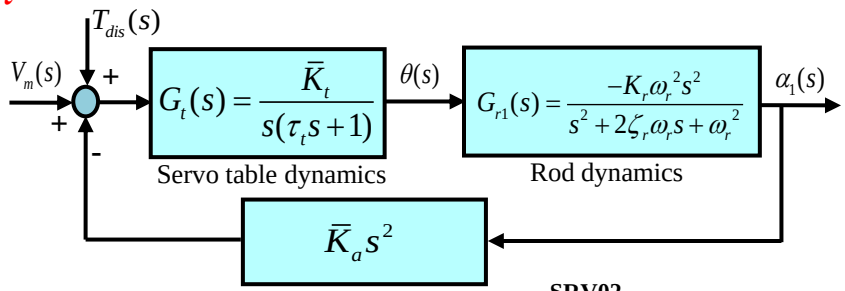
$$G_t(s) = \frac{\Omega(s)}{u(s)} = \frac{\bar{K}_t}{\tau_t s+1}$$

$$\frac{\alpha(s)}{\theta(s)} = \frac{K_r \omega_r^2 s^2}{s^2 + 2\zeta_r \omega_r s - \omega_r^2}$$

$$\theta(s) = G_t(s)[u(s) + T_{dis}(s) + \bar{K}_a s^2 \alpha(s)]$$

$\alpha_1 = \alpha - \pi$

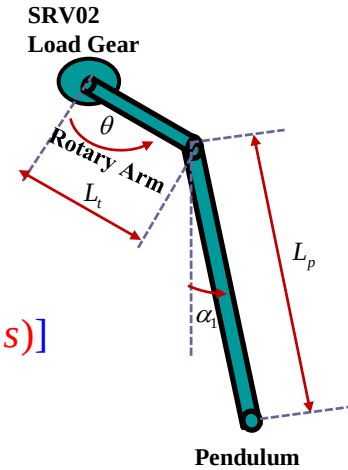
- **Rod downward**



$$G_t(s) = \frac{\Omega(s)}{u(s)} = \frac{\bar{K}_t}{\tau_t s+1}$$

$$G_{r1}(s) = \frac{\alpha_1(s)}{\theta(s)} = \frac{-K_r \omega_r^2 s^2}{s^2 + 2\zeta_r \omega_r s + \omega_r^2}$$

$$\theta(s) = G_t(s)[u(s) + T_{dis}(s) - \bar{K}_a s^2 \alpha_1(s)]$$



Unknown parameters: $K_r, \omega_r^2, \zeta_r, K_t, \tau_t, K_a, A_c$

Parameters of Nominal System

➢ **Physical parameters of nominal model**

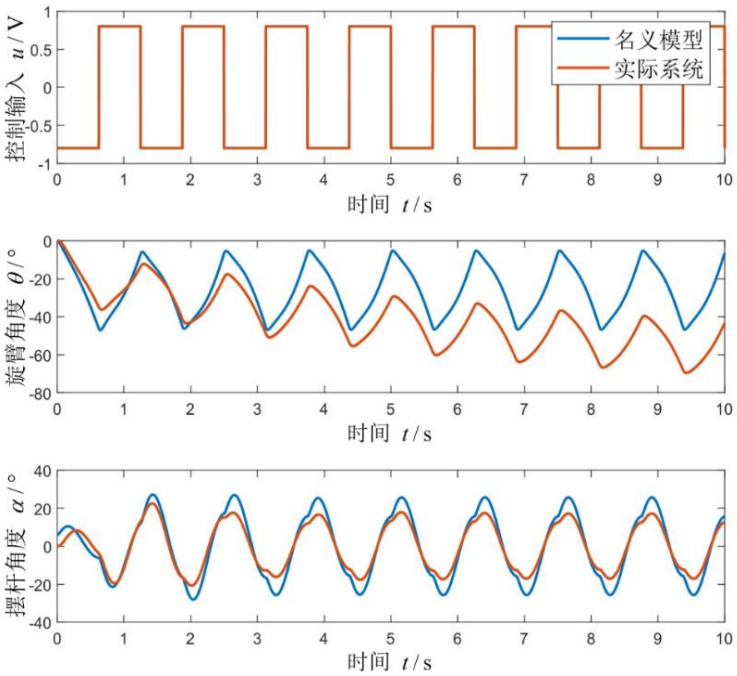
Symbol	Description	Value	Unit	Memo
m_p	Mass of pendulum	0.127	kg	Difficult to determine
L_c	Half of the total length of pendulum	0.1685	m	
L_p	Total length of pendulum	0.337	m	
J_p	Pendulum moment of inertia about center of mass	0.0012	kg.m^2	Difficult to determine
B_p	viscous damping coefficient of the pendulum	0.0024	N.m.s/rad	Difficult to determine
L_t	Rotary arm length from pivot to tip	0.216	m	
J_{eq}	Rotary arm moment of inertia about its center of mass	0.001	kg.m^2	Difficult to determine
B_{eq}	Rotary arm viscous damping coefficient as seen at the pivot axis	0.0024	N.m.s/rad	Difficult to determine
R_m	Motor armature resistance	2.6	Ω	
k_t	Motor current-torque constant	7.68×10^{-3}	N.m/A	
k_m	Motor back-EMF constant	7.68×10^{-3}	V/(rad/s)	
K_g	Total gear ratio	70		
η_m	Motor efficiency	1		
η_g	Gearbox efficiency	1		
V_m	Applied voltage to the motor	-10~10	V	
A_c	Coefficient of Coulomb friction torque	0.2?	V	Difficult to determine

Nominal parameters

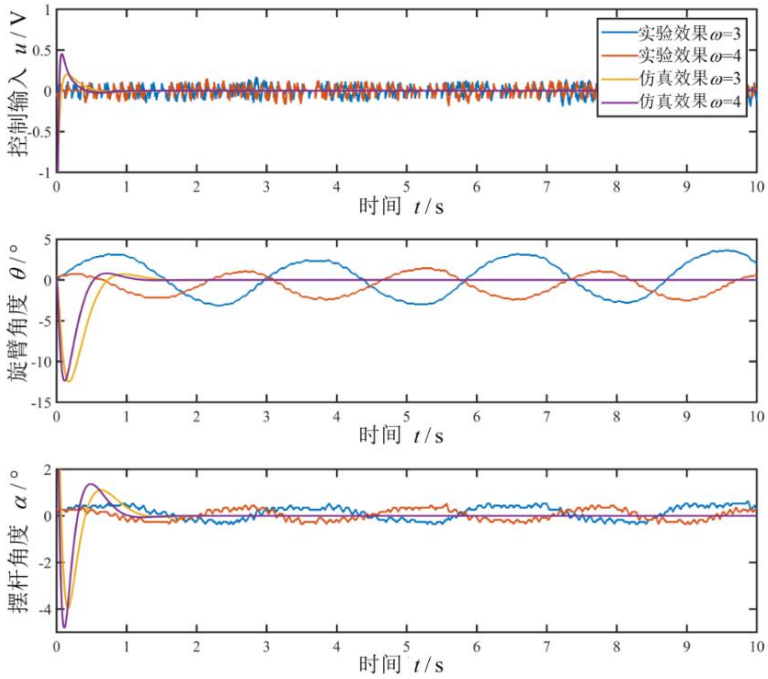
$$\bar{K}_t = 1.8208$$
$$\tau_t = 0.0610$$
$$A_c = 0.2$$
$$K_a = 0.0224$$
$$\omega_r = 6.6059$$
$$\xi_r = 0.0378$$
$$K_r = 0.022$$

True value of real model?

Comparison of Experiment and Simulation



Comparison of open-loop control of rotary pendulum



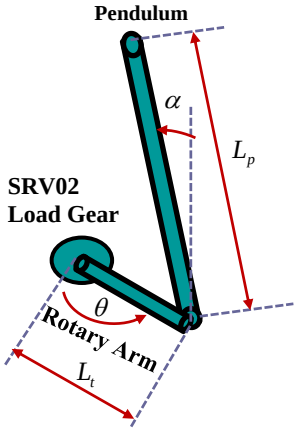
Comparison of closed-loop control of rotary pendulum

It's necessary to conduct system identification for decreasing the model error and possibly improving the control performance!

System Identification of Inverted Pendulum

- **Transfer function from input voltage to the table angle**
  - **Rod upward**

$$\theta(s) = G_t(s)[u(s) + T_{dis}(s) + \bar{K}_a s^2 \alpha(s)]$$



  - **Rod downward**

$$\theta(s) = G_t(s)[u(s) + T_{dis}(s) - \bar{K}_a s^2 \alpha_1(s)]$$

I/O model in time domain

$$\tau_t \ddot{\theta} + \dot{\theta} = \bar{K}_t u(t) - \bar{K}_t A_c \mathrm{sign}(\dot{\theta}) - \bar{K}_t \bar{K}_a \ddot{\alpha}_1$$

LSE in time domain

  - **Regressor model**

$$y_1 = \varphi_1^T \beta_1$$

Exiting the system and Identical filter for each channel

$$y_{f1} = \varphi_{1f}^T \beta_1$$

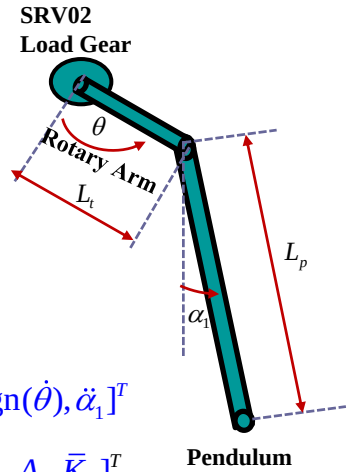
Parameter estimation with LSE

$$\hat{\beta}_1 = (\varphi_{1f}^T \varphi_{1f})^{-1} \varphi_{1f}^T y_{1f}$$

$$y_1 = u$$

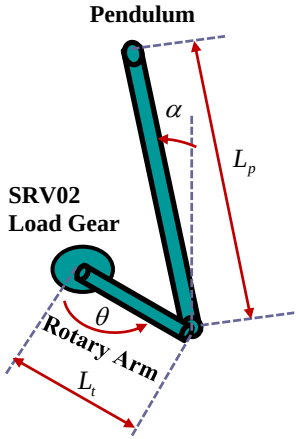
$$\varphi_1 = [\ddot{\theta}, \dot{\theta}, \mathrm{sign}(\dot{\theta}), \ddot{\alpha}_1]^T$$

$$\beta_1 = [\frac{\tau_t}{\bar{K}_t}, \frac{1}{\bar{K}_t}, A_c, \bar{K}_a]^T$$



System Identification of Inverted Pendulum System

- **Transfer function from table angle to rod angle**



➢ **Rod upward**

$$G_r(s)=\frac{\alpha(s)}{\theta(s)}=\frac{K_r\omega_r^2 s^2}{s^2+2\zeta_r\omega_r s-\omega_r^2}$$

VS

➢ **Rod downward**

$$G_{r1}(s)=\frac{\alpha_1(s)}{\theta(s)}=\frac{-K_r\omega_r^2 s^2}{s^2+2\zeta_r\omega_r s+\omega_r^2}$$

I/O model in time domain

$$-L_t/g\cdot\omega_r^2\ddot{\theta}=\ddot{\alpha}_1+2\zeta_r\omega_r\dot{\alpha}_1+\omega_r^2\alpha_1$$

LSE in time domain

➢ **Regressor model**

$$y_2=\varphi_2^T\beta_2$$

$$y_2=\ddot{\alpha}_1$$

$$\varphi_2=[-\ddot{\theta},-\dot{\alpha}_1,-\alpha_1]^T$$

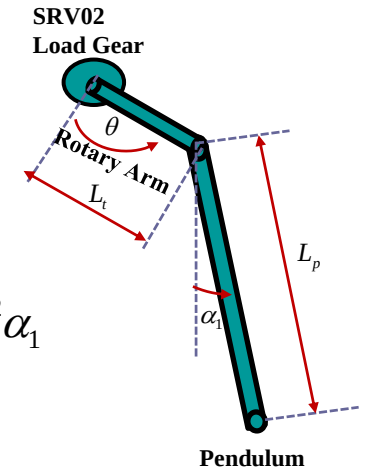
$$\beta_2=[L_t/g\cdot\omega_r^2,2\zeta_r\omega_r,\omega_r^2]^T$$

Exciting system and Identical filter for each channel

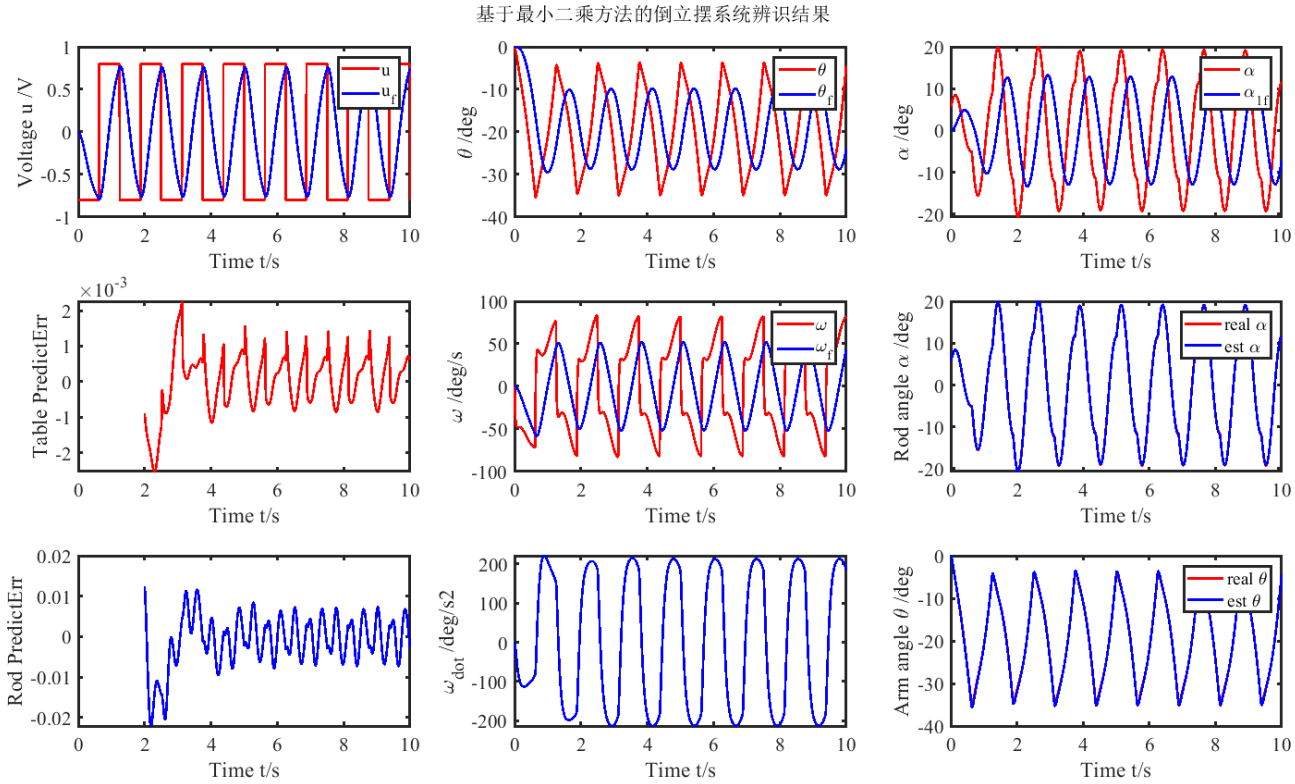
$$y_{2f}=\varphi_{2f}^T\beta_2$$

➢ **Parameter estimation**

$$\hat{\beta}_2=(\varphi_{2f}^T\varphi_{2f})^{-1}\varphi_{2f}^T y_{2f}$$



Simulation Results of System Identification



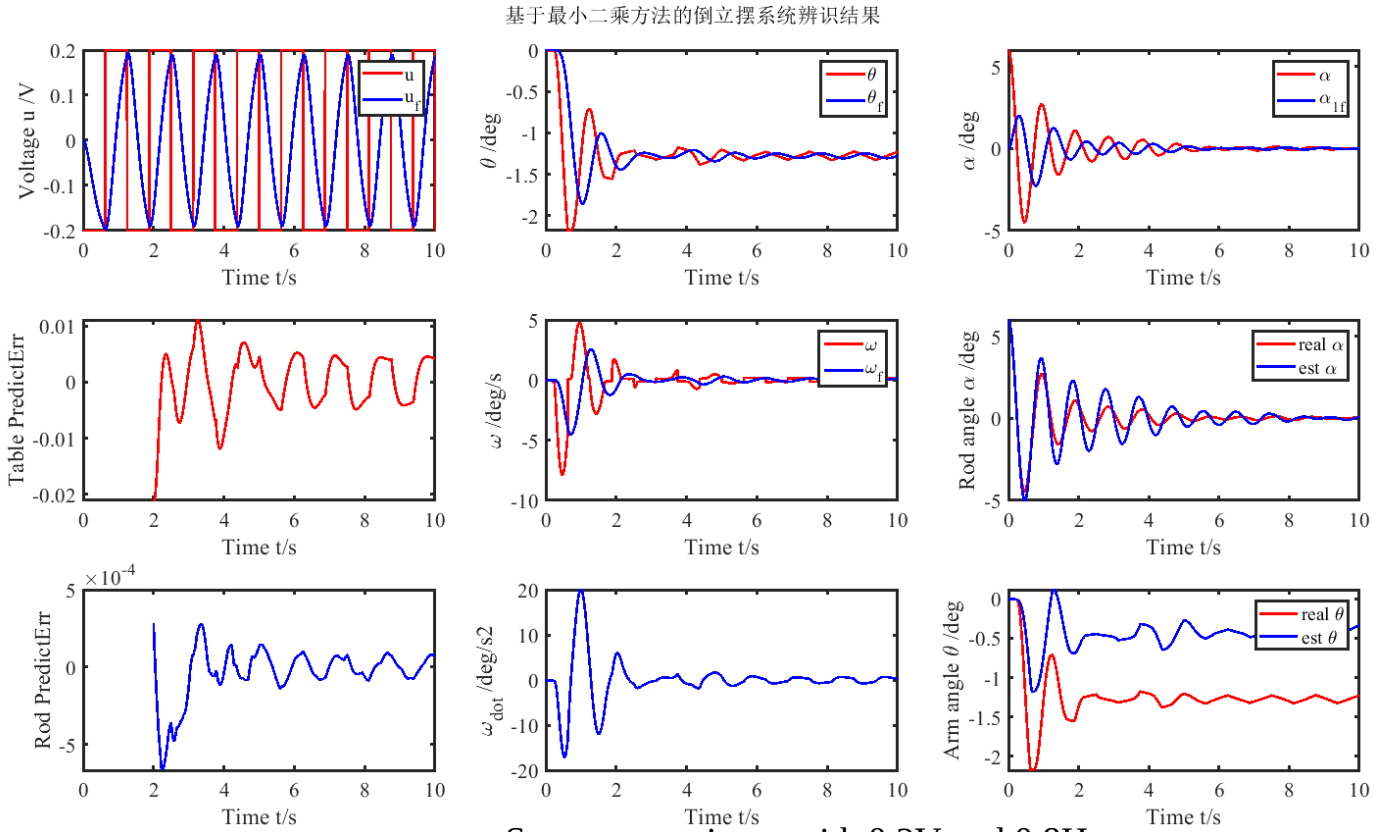
Square wave input with 0.8V and 0.8Hz

u=0.8V,σ^2=0deg^2

Parameter estimate

$$\bar{K}_t = 1.5895$$
$$\tau_t = 0.0595$$
$$A_c = 0.1090$$
$$K_a = 0.0241$$
$$\omega_r = 6.5854$$
$$\xi_r = 0.0407$$
$$K_r = 0.0216$$
$$\Delta_\theta = -0.0014$$

Simulation Results of System Identification



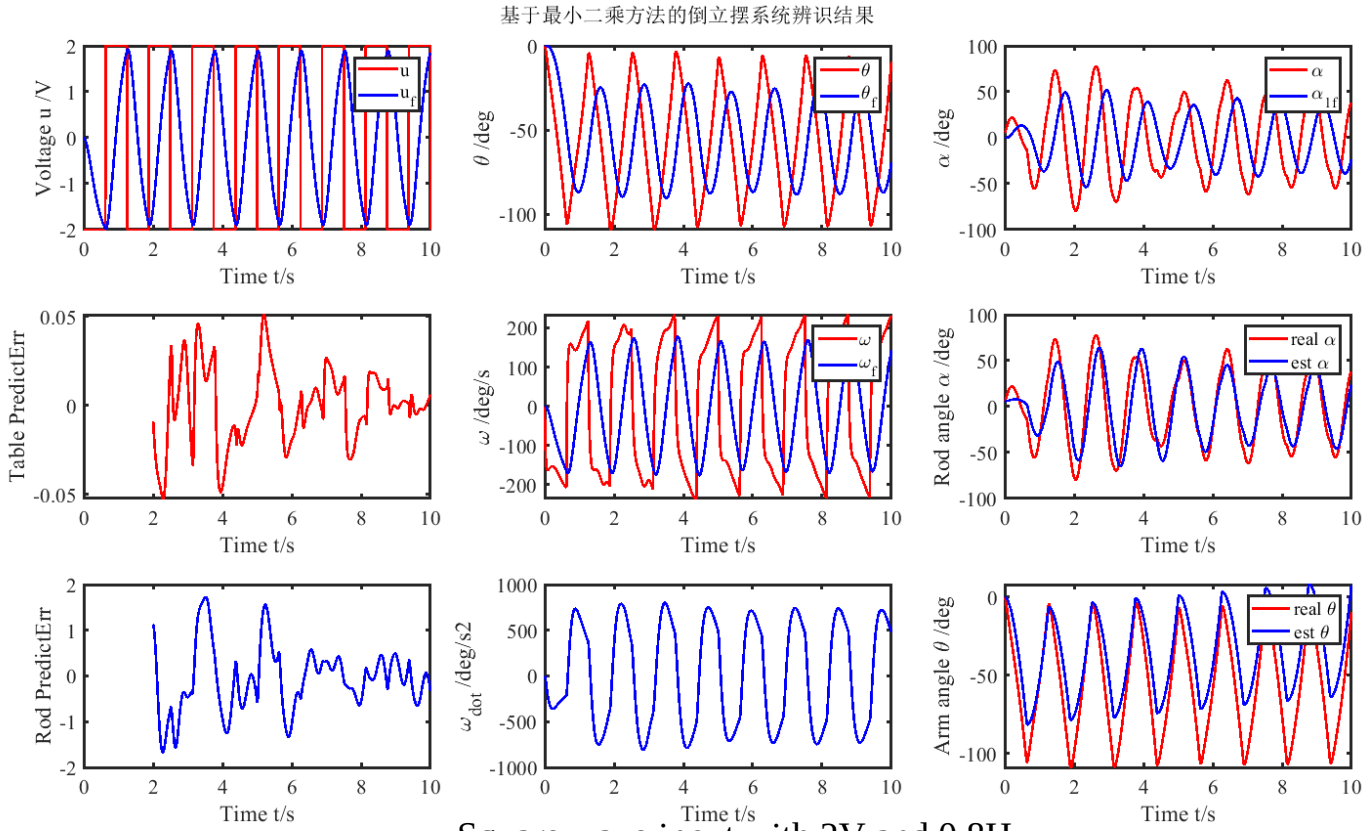
Square wave input with 0.2V and 0.8Hz

u=0.2V,σ^2=0deg^2

Parameter estimate

$$\bar{K}_t = 1.2877$$
$$\tau_t = 0.4213$$
$$A_c = 0.1955$$
$$K_a = 0.0339$$
$$\omega_r = 6.6310$$
$$\xi_r = 0.0401$$
$$K_r = 0.0219$$
$$\Delta_\theta = -0.00049$$

Simulation Results of System Identification



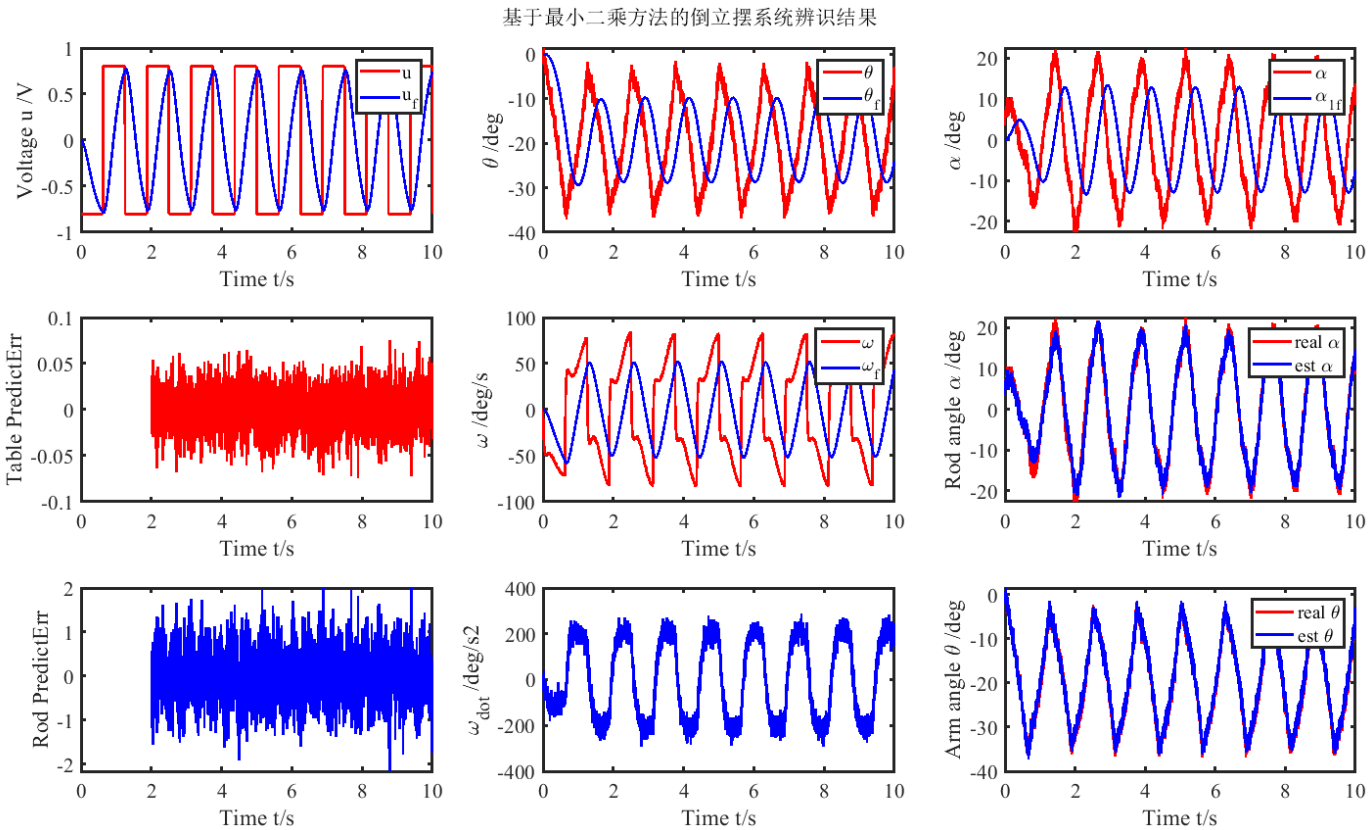
Square wave input with 2V and 0.8Hz

u=2V,σ^2=0deg^2

Parameter estimate

$$\bar{K}_t = 19.131$$
$$\tau_t = 0.5328$$
$$A_c = 1.7153$$
$$K_a = 0.0017$$
$$\omega_r = 6.1447$$
$$\xi_r = 0.0516$$
$$K_r = 0.0164$$
$$\Delta_\theta = -0.0047$$

Simulation Results of System Identification



Square wave input with 0.4V and 0.8Hz

u=0.8V,σ^2=1deg^2

Parameter estimate

$$\bar{K}_t = 1.4185$$
$$\tau_t = 0.0529$$
$$A_c = 0.0455$$
$$K_a = 0.0289$$
$$\omega_r = 6.1157$$
$$\xi_r = 0.0291$$
$$K_r = 0.0166$$
$$\Delta_\theta = -0.0017$$

Simulation Results of System Identification

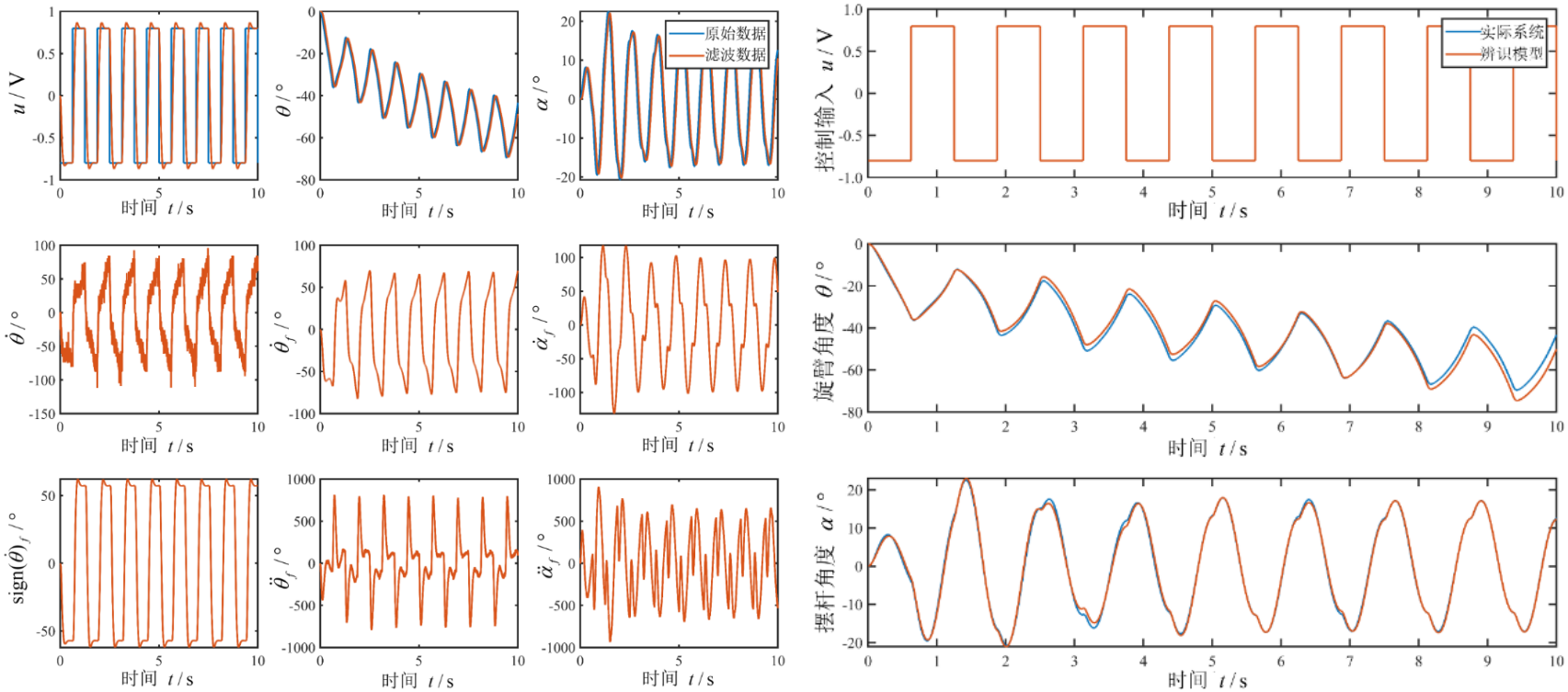
Simulation with different exciting signals

Parameter estimate	σ^2=0deg^2				σ^2=1deg^2				True value
	0.2V	0.8V	1.0V	2V	0.2V	0.8V	1.0V	2V	
K_t	1.2877	1.5895	1.9771	19.131	0.1563	1.4185	1.5707	57.7306	1.8208
τ_t	0.4213	0.0595	0.0667	0.5328	0.0012	0.0529	0.0561	1.7588	0.0610
A_c	0.1955	0.1090	0.2445	1.7153	1.8199	0.0455	0.0820	1.8387	0.2
K_a	0.0339	0.0241	0.0188	0.0017	0.0058	0.0289	0.0249	0.0010	0.0224
ω_r	6.6310	6.5854	6.5506	6.1447	6.3477	6.1157	6.2555	6.1213	6.6059
ζ_r	0.0401	0.0407	0.0427	0.0516	0.1252	0.0291	0.0350	0.0507	0.0378
K_r	0.0219	0.0216	0.0214	0.0164	-0.0001	0.0166	0.0183	0.0161	0.0220
	-0.00049	-0.0014	-0.0018	-0.0047	-0.0085	-0.0017	-0.0018	-0.0097	0

Inappropriate exciting signals for the system

What will the parameter estimate be effected by the input disturbances and measurement ?

Experimental Results of System Identification

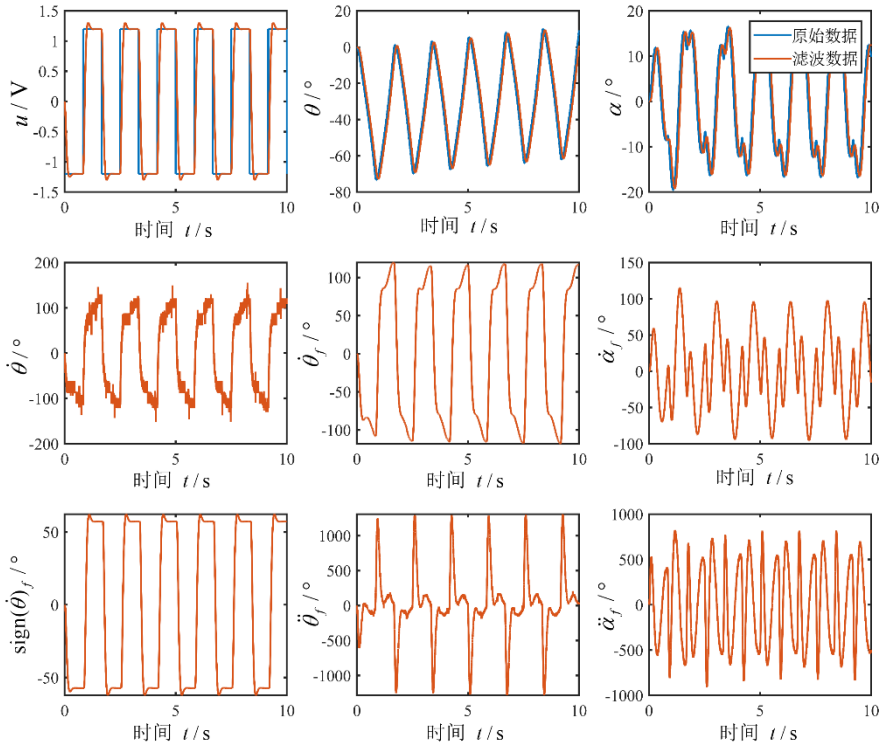


Exciting signals and corresponding filtered signals

Model validation

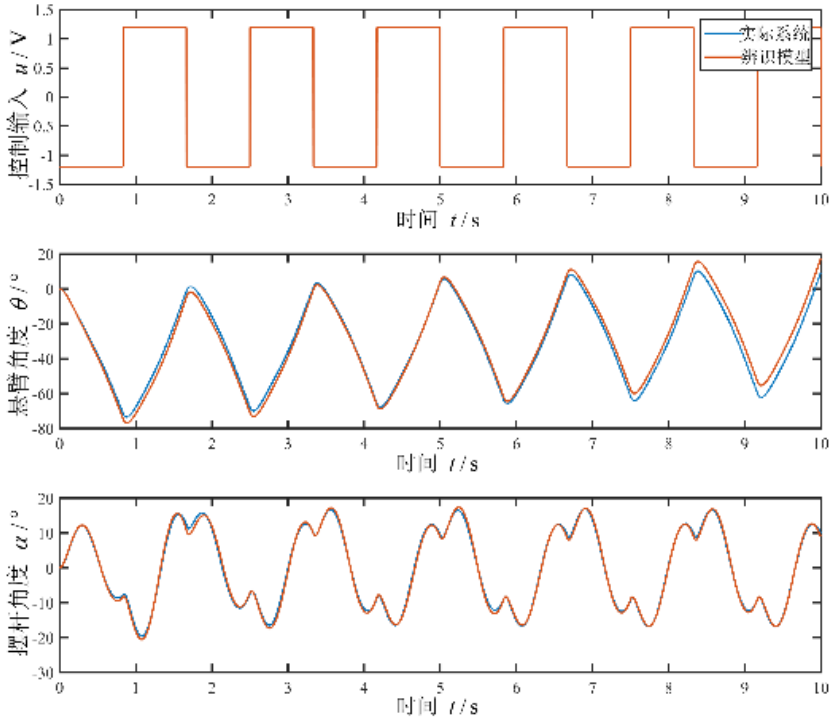
Square wave with 0.8V and 0.8Hz

Experimental Results of System Identification



Exciting signals and corresponding filtered signals

Square wave with 1.2V and 0.6Hz



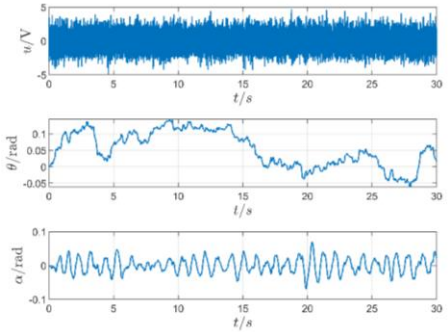
Model validation

Experimental Results of System Identification

- **Comparison of experimental results with different exciting signals**

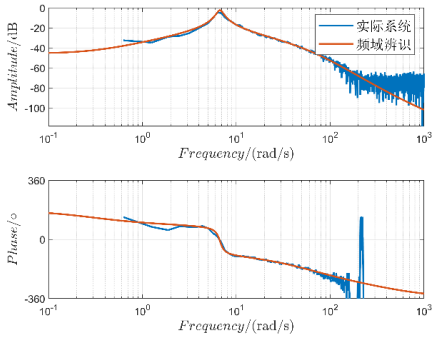
Parameters/residuals	Nominal value	0.8V-0.8Hz LSE	1.2V-0.6Hz LSE
τ_t	0.0612	0.1468	0.1342
K_t	1.8208	2.7546	2.6553
A_c	0	0.3578	0.4924
K_a	0.0224	0.0136	0.0111
L_t	0.2160	0.2084	0.2154
ω_r	6.6059	6.6162	6.5049
ζ_r	0.0378	0.0007	-0.0136
Δ_1	0	0.0258	-0.01364
RMS$\lvert e_u \rvert$		0.0163	0.0164
RMS$\lvert e_{\ddot{\alpha}} \rvert$		5.5328	8.5202

Experimental Frequency Identification of Inverted Pendulum

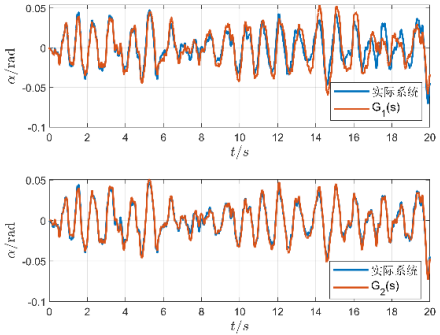


Exciting and output signals in time domain

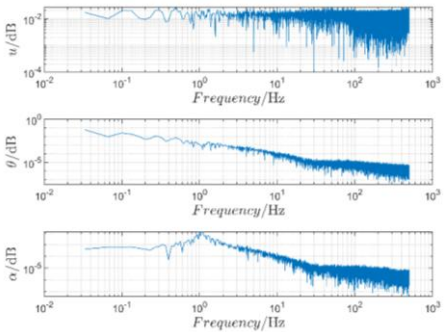
Exciting signal: sum of sinusoids with *Amplitude* =5, *frequency*=[0.5,500]Hz, *N*=10000



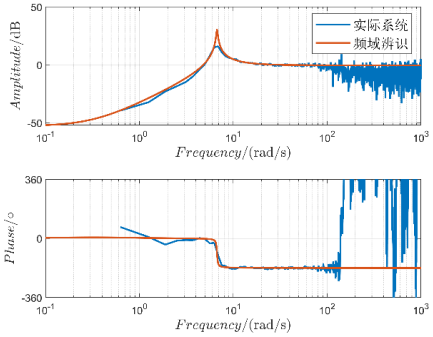
$$G_2(s)=\frac{\alpha_1}{u}=\frac{-8.903s^2+3371s-1641}{s^4+141.6s^3+3970s^2+10490s+168100}$$



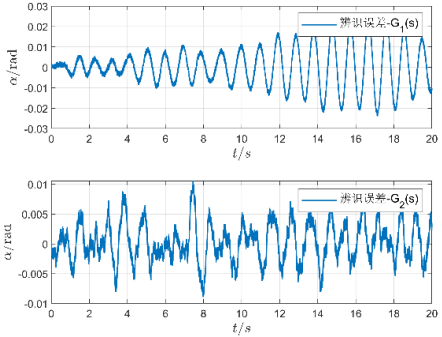
Actual output and simulation output



Exciting and output signals in time domain



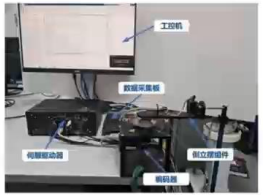
$$G_1(s)=\frac{\alpha_1}{\theta}=\frac{-0.9475s^2}{s^2+0.2115s+43.03}$$



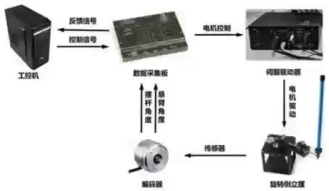
Output error

Effect of System Identification in Feedback Control

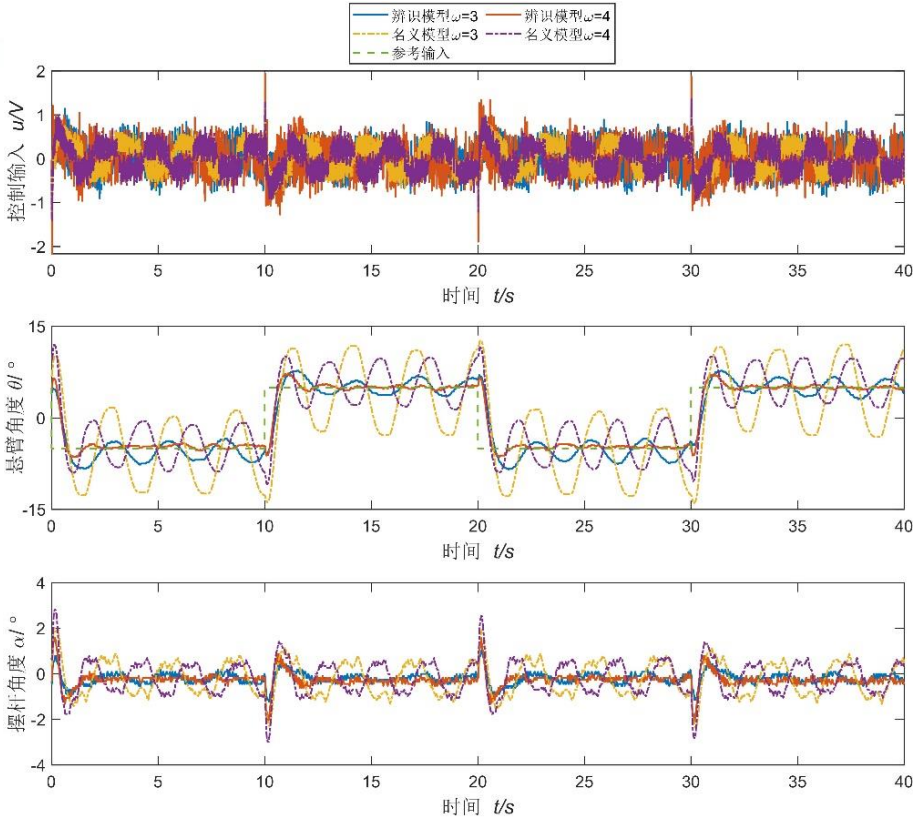
旋转倒立摆系统的辨识及控制演示视频



旋转倒立摆实验系统



旋转倒立摆工作原理示意



State feedback control with nominal model and identified model

The swing angle of the inverted pendulum is smaller with identified model than that with nominal model!